

BikeMap UB системийн ерөнхий архитектур

Client

Browser
(desktop)

Browser
(mobile)

Admin browser

Strava / Garmin GPX export

Frontend

Frontend — Django Templates + Vanilla JavaScript

Bootstrap 5
(CDN)

Leaflet
L.map / L.marker

auth.js
login / logout

api.js
fetch wrapper

gps.js / segment.js
poi.js / route.js

API & Auth

API & Auth Layer — Django + DRF

JWT Cookie Auth
httpOnly, SameSite=Strict

RBAC Permissions
guest/cyclist/mod/admin

Throttle
login 10/min, anon 60/min

drf-spectacular
/api/schema | /api/docs (Swagger)

Security middleware: X-Frame=DENY | HSTS | XSS-filter | CONTENT-NOSNIFF | CSRF

OSM tile direct
(client → tile server)

Backend Apps

Backend Apps — Django app modules

apps/
accounts

JWT, RBAC,
profile, password reset

apps/
segments

CRUD segment,
condition tagging

apps/
pois

POI CRUD, vote,
approve/reject

apps/
aggregation

update_aggregation()
spatial hash + argmax

apps/
routes

Smart Route,
GPX import / export

apps/
audit_log

Admin actions
log (IP-тэй)

Data Layer

PostgreSQL (production) / SQLite (dev)

User

Segment

POI

POIVote

Crowd
Aggregation

Audit
Log

Reset
Token

External Services

OSRM
cycling profile
router.project
-osrm.org

OpenStreetMap
Tile Server
(map.tile)

Infra

Infra / Deploy:

Gunicorn
WSGI

WhiteNoise
static

Nginx
proxy

Docker
Compose

GitHub
Actions CI

Heroku
/ VPS